

Trapping Light in Diamond: The NV Center as a Room-Temperature Qubit

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DFT confirms NV⁻ as the thermodynamically stable, spin-triplet charge state in diamond, with a computed ZPL consistent with optical spin-readout for quantum sensing.

Why Diamond?

- Superconducting qubits need near-zero cooling — costly, hard to scale
- Point defects in wide-bandgap crystals offer a room-temperature alternative
- NV center forms when, inside diamond's carbon lattice, nitrogen replaces one carbon atom and a carbon is removed, leaving an vacancy.

NV center is a spin qubit that can as qubit and nanoscale sensor, this project focuses on the properties enabling the optical spin-readout.

Requirement	Why it matters	Method
Optically active	Can interact with light — enables initialization and readout	ZPL calculation
Highly localized states	Individually addressable — a single, well-defined qubit	LDOS
Levels inside the band gap	Isolated from bulk bands — accessible, distinguishable transitions	Formation energy
Paramagnetic ground state	Nonzero spin — enables spin-based control and readout	Kohn-Sham diagram
Thermodynamically stable	Ensures the defect actually forms and persists in that charge state	Formation energy

Building the Defect, Step by Step

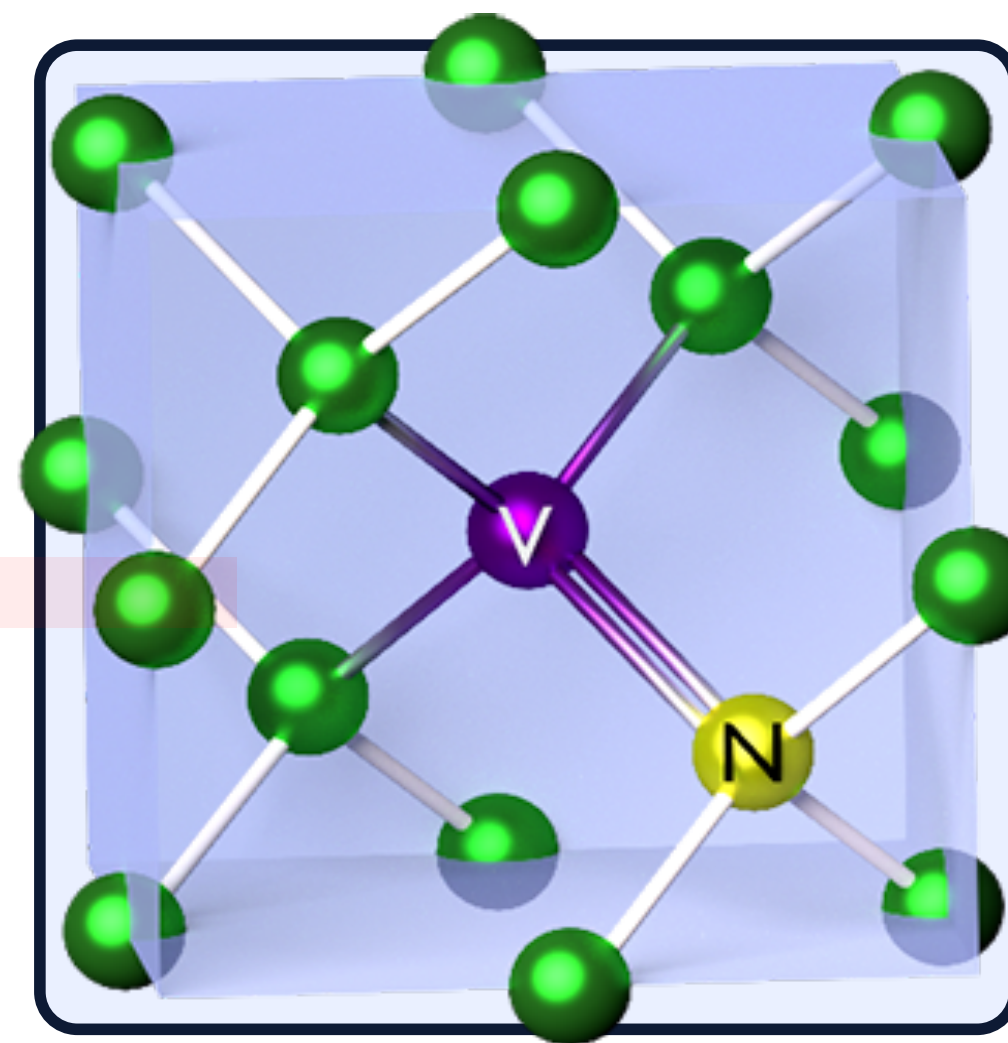


Figure 1: NV center in diamond

- 1 Bulk Validation**
Evaluate lattice constant
PBE vs. HSE06
- 2 Defect Construction**
Build NV supercell
(N substitution + C vacancy)
- 3 Density of States**
Identify in-gap defect states
Localization
- 4 Kohn-Sham States**
Determine level occupations
- 5 Thermodynamics**
Calculate formation energies
Transitions levels
- 6 Optical Properties**
Compute zero phonon line
Wavelength

Which Charge State Wins?

To find the thermodynamically favorable charge state, we compute NV's formation energy across all charges ($q = -3$ to $+2$) vs. Fermi level:

$$E = E_{\text{def}}^q - E_{\text{perf}} + 2\mu_C - \mu_N + q(E_{\text{VBM}} + E_F) + E_{\text{corr}}$$

total energies
defect vs
perfect cell

atomic
chemical
potentials

electron
reservoir
term

finite-size
charge
correction

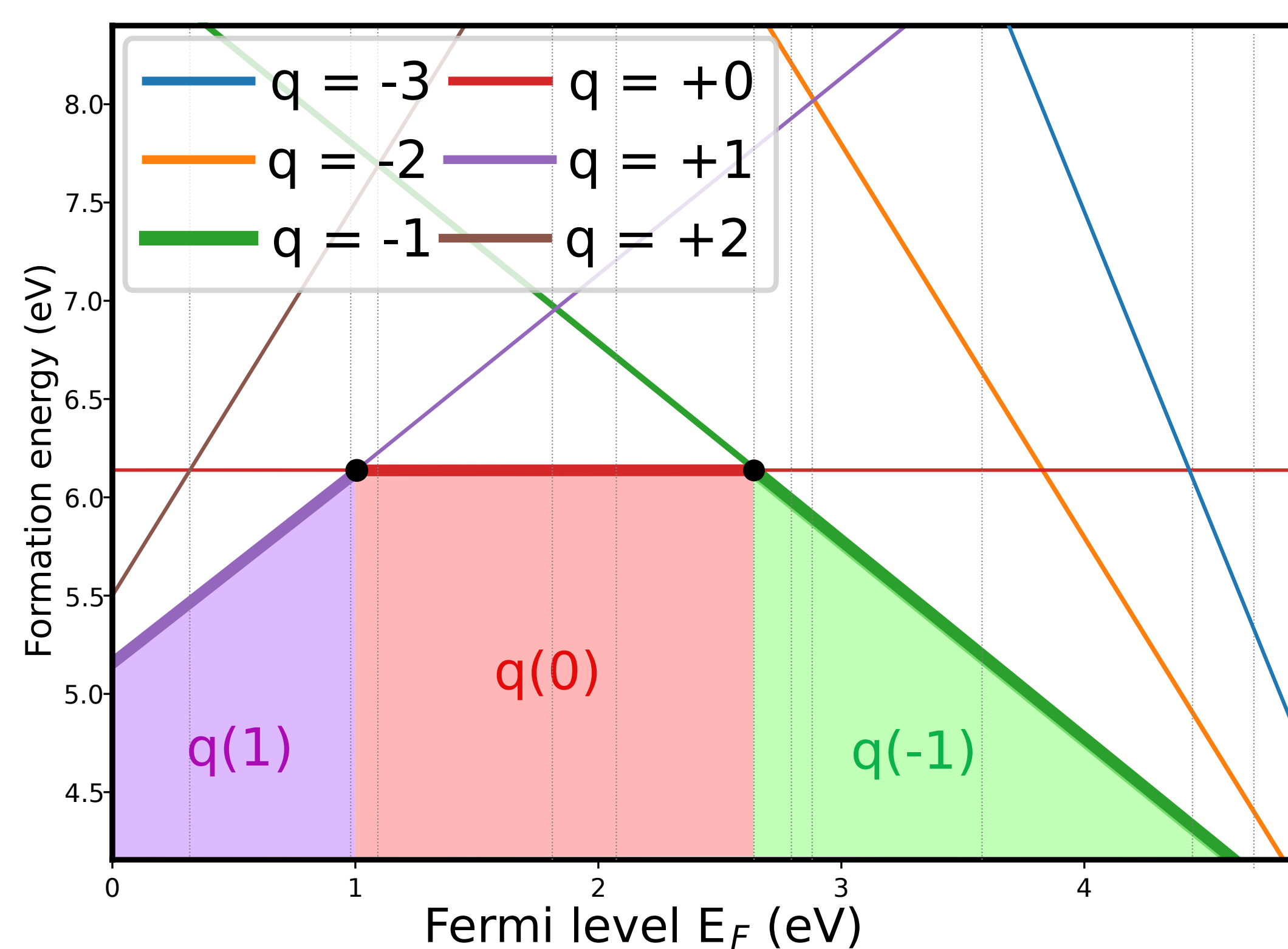


Figure 2: NV⁻ is the lowest-formation-energy charge state across most of the gap ($E_F > 2.64$ eV) — including the range typical of N-doped diamond — confirming it as the thermodynamically stable.

How NV⁻ Talks to Light

Three charge states dominate, $q = +1, 0, -1$, with spin and occupation determining which is qubit-relevant.

Charge	Spin (S)	Unpaired e
+1	0.0	0
0	0.5	1
-1	1.0	2

Table 1: NV⁻ is the only relevant charge state with a spin-triplet $S=1$ ground state, two unpaired electrons make optical spin-readout possible.

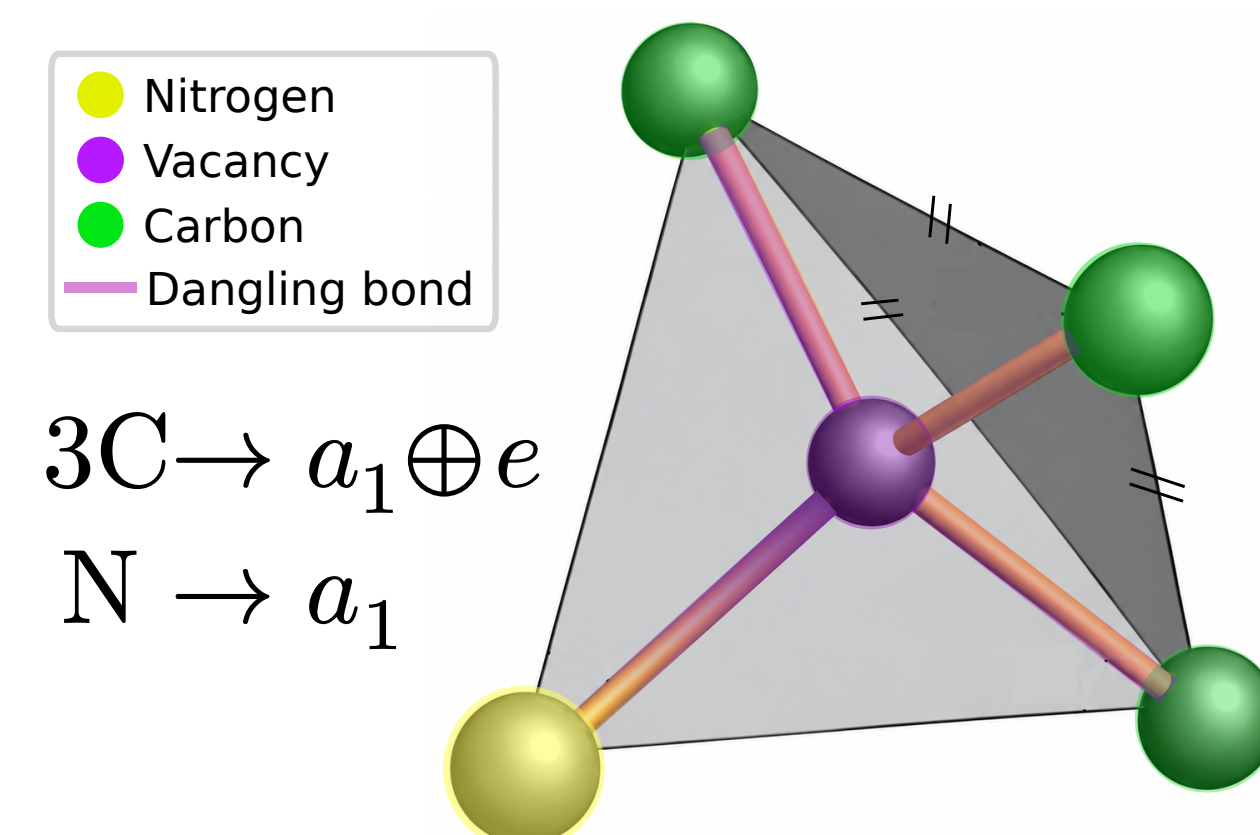


Figure 3: 120° rotation about the N-vacancy axis (C_{3v}) maps the three C dangling bonds onto each other, splitting them into a_1 and e .

Figure 4: One electron moves from a_1 (filled) to the doubly-degenerate e level, driving the ${}^3A_2 \rightarrow {}^3E$ optical transition

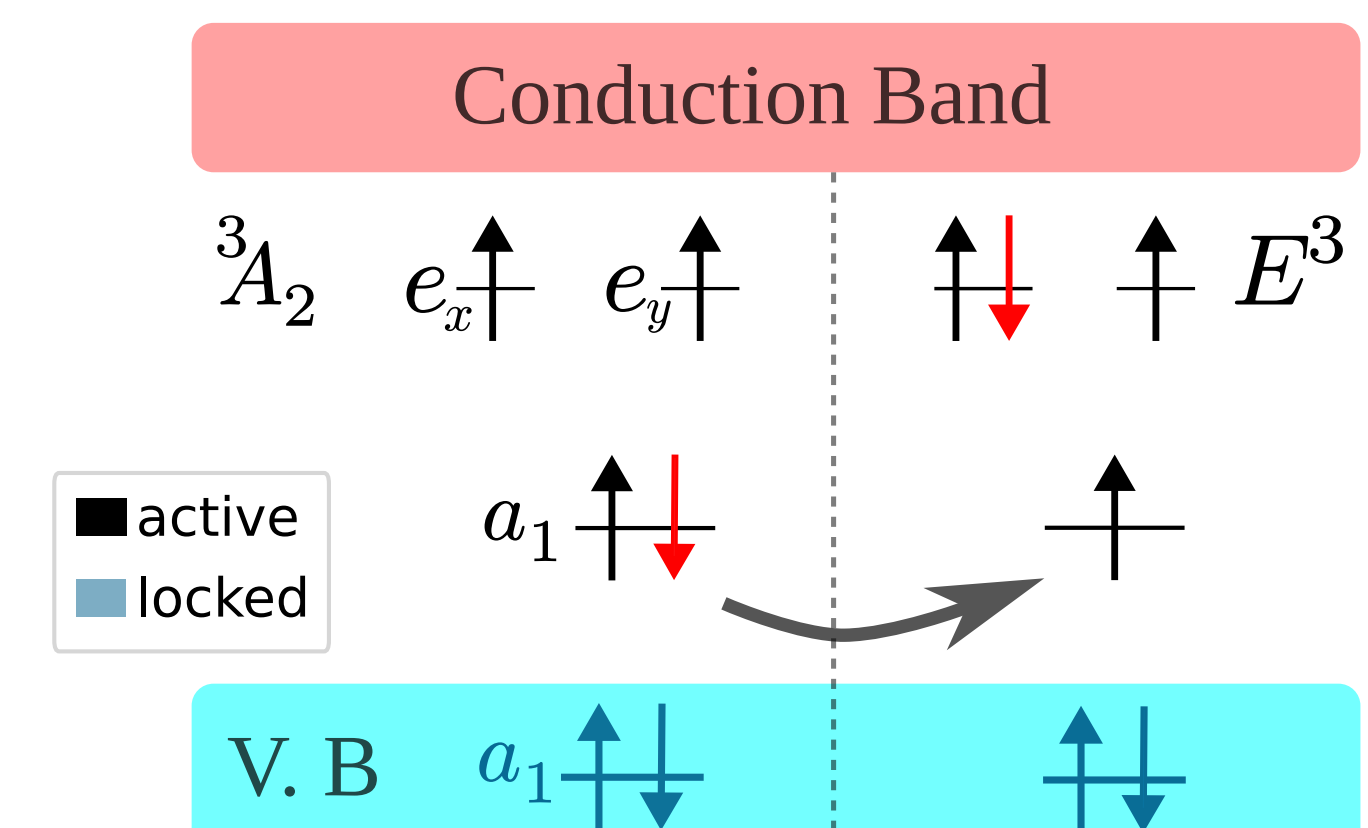
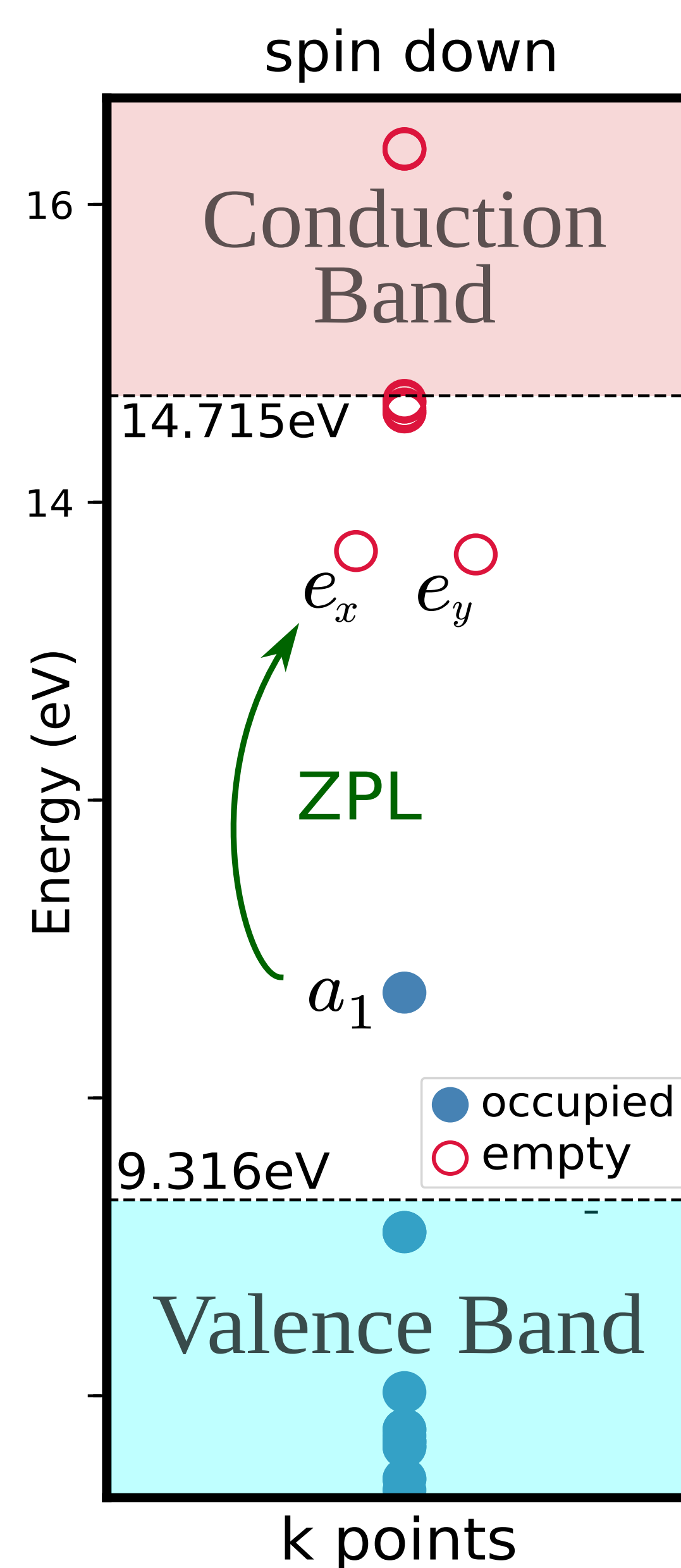


Figure 5: Of the two a_1 combinations from C_{3v} symmetry, one merges into the valence band (locked, always filled); the other remains in-gap, active alongside the e pair.

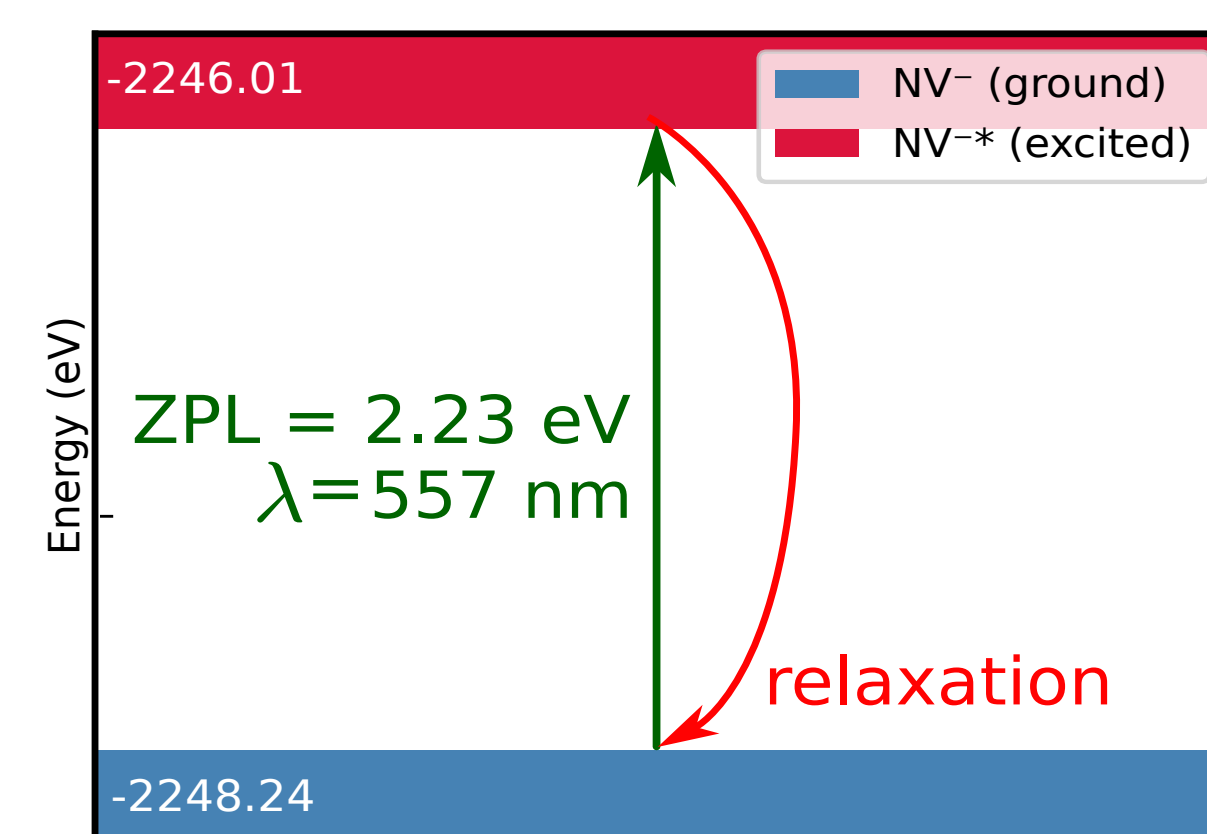


Figure 6: Computed ZPL lies in the visible spectrum, confirming NV's optical transition can be addressed with visible light, basis

Can We Trust the Method?

Quantity	PBE	HSE06	Exp
Lattice constant (Å)	3.5737	3.5481	3.567
Band gap (eV)	4.12	5.34	5.47
ZPL (eV)		2.23	1.945

Table 2: HSE06 corrects PBE's underestimation, justifying its use for defect calculations

Quantity (eV)	This work	Literature
transition level $\epsilon(0+)$	0.981	0.9–1.2
transition level $\epsilon(0/-)$	2.64	≈2.7–2.8

Table 3: Formation energies depend on the chemical potential reference used and are best compared within the same convention.

Confirm the DFT setup reproduces known diamond physics and that the NV-specific quantities are internally consistent.

What This Means For Quantum Sensing

NV⁻ is both thermodynamically stable and optically addressable — properties already exploited in real-world NV-diamond quantum sensors — and this work provides first-principles confirmation of why NV⁻ is the charge state responsible.

- Point defects like the NV center are emerging as a strong platform bridging quantum sensing, computation, and scalable room-temperature devices, future works includes long coherence time calcs.

References

- Dreyer et al., Annu. Rev. Mater. Res. 48, 1 (2018).
- Panana Vera, J. I., Thesis, UNMSM
- Doherty et al., Phys. Rep. 528, 1 (2013).

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